

React HandsOn 16:-

Theory:

- Explain react form validation.
React Forms Validation is the process of checking whether the user input satisfies predefined rules before the form is submitted. Validation helps ensure that the data entered is accurate, complete, and in the correct format. In React, validation is usually implemented using event handlers such as `onChange` and `onSubmit`, along with the component's state. If any validation rule fails, appropriate error messages are displayed to the user, preventing invalid data from being processed.
- Identify the difference between the react form and the html form.

React Form	HTML Form
Form data is managed using React state.	Form data is managed directly by the browser.
Uses controlled components for better control over inputs.	Input elements manage their own state.
Validation is implemented using JavaScript and React event handlers.	Form submission reloads the page by default.
Provides dynamic updates without refreshing the page.	Usually refreshes or navigates to another page after submission.

- Explain about controlled components.

A Controlled Component is a form element whose value is controlled by the React state. The value displayed in the input field is stored in the component's state and updated through the `onChange` event. This allows React to have complete control over the form data, making validation and dynamic updates easier to implement.

```
<input
  type="text"
  name="name"
  value={this.state.name}
  onChange={this.handleChange}
/>
```

- Identify various React Form input controls.

React supports all standard HTML form controls, including:

1. Text Box (`<input type="text">`) – Used for entering single-line text such as names.
 2. Password (`<input type="password">`) – Used for securely entering passwords.
 3. Email (`<input type="email">`) – Used for entering email addresses.
 4. Textarea (`<textarea>`) – Used for multi-line text input such as comments or feedback.
 5. Checkbox (`<input type="checkbox">`) – Used for selecting one or more options.
 6. Radio Button (`<input type="radio">`) – Used for selecting one option from a group.
 7. Dropdown (`<select>`) – Used for choosing an option from a list.
 8. Button (`<button>`) – Used to submit forms or trigger actions.
- Explain how to handle react form.

Handling forms in React involves storing user input in the component's state and updating it whenever the user types or interacts with the form. The `onChange` event is used to capture input changes, while `setState()` (or `useState()` in functional components) updates the stored values. This approach keeps the form inputs synchronized with the component's state and allows easy validation and processing of data.

- Explain about submitting a form in react.

In React, forms are submitted using the `onSubmit` event of the `<form>` element. The event handler usually calls `event.preventDefault()` to stop the browser from reloading the page. The form data is then validated, processed, or sent to a server. If the validation succeeds, the required action—such as displaying a success message, storing the data, or making an API call—is performed. This approach provides a smooth user experience without refreshing the page.

Practical:-

Code:

```
# App.css ticketraisingapp\... M JS index.js ticketraisingapp\... M JS register.js U X JS App.js mailregisterapp\... M # App.css mailregisterapp\... M JS index.js mailregisterapp\... M
1 import React, { Component } from "react";
2
3 class Register extends Component {
4   constructor(props) {
5     super(props);
6     this.state = {
7       name: "",
8       email: "",
9       password: "",
10      errors: {}
11    };
12  }
13   handleChange = (event) => {
14     this.setState({
15       [event.target.name]: event.target.value
16     });
17   };
18   validateForm = () => {
19     let errors = {};
20     if (this.state.name.length < 5) {
21       errors.name = "Name must contain at least 5 characters";
22     }
23     if (
24       !this.state.email.includes("@") ||
25       !this.state.email.includes(".")
26     ) {
27       errors.email = "Enter a valid email address";
28     }
29     if (this.state.password.length < 8) {
30       errors.password = "Password must contain at least 8 characters";
31     }
32     this.setState({ errors });
33     return Object.keys(errors).length === 0;
34   };
35 }
```

```

36   handleSubmit = (event) => {
37     event.preventDefault();
38     if (this.validateForm()) {
39       alert("Registration Successful");
40       this.setState({
41         name: "",
42         email: "",
43         password: "",
44         errors: {}
45       });
46     }
47   };
48   render() {
49     return (
50       <div className="container">
51         <h2>Mail Registration</h2>
52         <form onSubmit={this.handleSubmit}>
53           <label>Name</label>
54           <input
55             type="text"
56             name="name"
57             value={this.state.name}
58             onChange={this.handleChange}
59           />
60           <p className="error">{this.state.errors.name}</p>
61           <label>Email</label>
62           <input
63             type="text"
64             name="email"
65             value={this.state.email}
66             onChange={this.handleChange}
67           />
68         </form>
69       </div>
70     );
71   }
72 }
```

```

66     />
67     <p className="error">{this.state.errors.email}</p>
68     <label>Password</label>
69     <input
70       type="password"
71       name="password"
72       value={this.state.password}
73       onChange={this.handleChange}
74     />
75     <p className="error">{this.state.errors.password}</p>
76     <button type="submit">
77       Register
78     </button>
79   </form>
80 </div>
81   );
82 }
83 }
84 export default Register;

```

```

# App.css ticketraisingapp\... M  JS index.js ticketraisingapp\... M  JS register.js U  JS App.js mailregisterapp\... M X  # App.css mailregister
1  import Register from "../register";
2  import "../App.css";
3
4  function App() {
5    return (
6      <div className="App">
7        <Register />
8      </div>
9    );
10 }
11
12 export default App;

```

```

1  .App{
2    display:flex;
3    justify-content:center;
4    align-items:center;
5    margin-top:40px;
6    font-family:Arial, Helvetica, sans-serif;
7  }
8
9  .container{
10   width:400px;
11   padding:20px;
12   border:1px solid gray;
13   border-radius:10px;
14   box-shadow:0px 0px 10px lightgray;
15 }
16
17 h2{
18   text-align:center;
19   color:navy;
20 }
21
22 input{
23   width:100%;
24   padding:8px;
25   margin-top:5px;
26   margin-bottom:5px;
27   box-sizing:border-box;
28 }

```

```

21
22   input{
23     width:100%;
24     padding:8px;
25     margin-top:5px;
26     margin-bottom:5px;
27     box-sizing:border-box;
28   }
29
30   button{
31     width:100%;
32     padding:10px;
33     background: green;
34     color: white;
35     border:none;
36     cursor:pointer;
37     font-size:16px;
38   }
39
40   button:hover{
41     background: darkgreen;
42   }
43
44   .error{
45     color: red;
46     font-size:14px;
47     margin:0 0 10px 0;
48   }

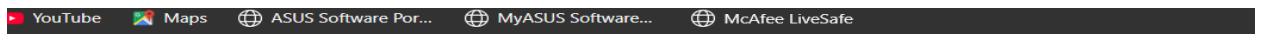
```

```

App.css ticketraisingapp\... M    JS index.js ticketraisingapp\... M    JS register.js U    JS App.js mailregisterapp\... M    # App.c
1  import React from "react";
2  import ReactDOM from "react-dom/client";
3  import "./index.css";
4  import App from "./App";
5
6  const root = ReactDOM.createRoot(document.getElementById("root"));
7
8  root.render(
9    <React.StrictMode>
10     | <App />
11     </React.StrictMode>
12  );
13

```

Output:

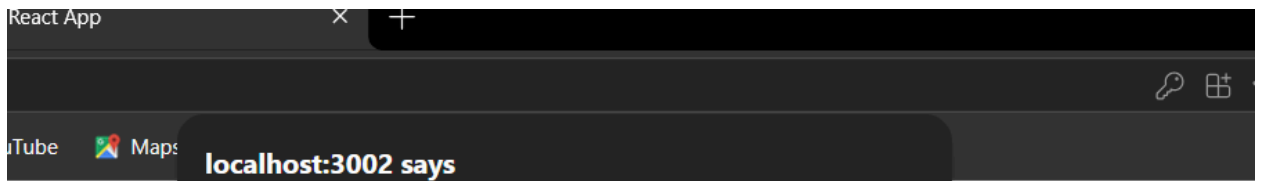


Mail Registration

Name

Email

Password



localhost:3002 says

Registration Successful

OK

Name

Riya Kumari

Email

nishant.22bce7457@vitapstudent.ac.in

Password

.....

Register